



PAPER ID-310601

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Subject Code: RCA1901

Roll No:

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MCA-INT
(SEM IX) THEORY EXAMINATION 2024-25
COMPUTER GRAPHICS

TIME: 3 HRS**M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A**2 x 07 = 14**1. Attempt *all* questions in brief.

a.	Define Refresh Buffer.
b.	Define Color-lookup Table.
c.	What do you mean by Geometric and Coordinate Transformation?
d.	Find the matrix that represents rotation of an object about the origin.
e.	What do you mean by 3D Transformation?
f.	Define Parametric Curves.
g.	List out various types of illumination models.

SECTION B**07 x 3 = 21**2. Attempt any *three* of the following:

a.	Discuss Raster scan and Random scan displays.
b.	Discuss 2D Scaling with respect to origin.
c.	Write short note on Rotation about an axis that is not parallel to one of the coordinate axes.
d.	Discuss various properties of B-spline curves.
e.	Discuss Depth Buffer Method of hidden surfaces.

SECTION C**07 x 1 = 7**3. Attempt any *one* part of the following:

a.	How you can define parallel and perpendicular lines. Discuss slope-Intercept line drawing algorithm.
b.	State and drive the steps involved in Bresenham's Circle drawing algorithm.

4. Attempt any *one* part of the following:**7 x 1 = 7**

a.	What do you mean by Concatenation of matrices? Express as a matrix (i.e. CTM) the transformation which magnifies an object about its center C(h,k).
b.	Write short note on Viewing Transformation.

5. Attempt any *one* part of the following:**07 x 1 = 7**

a.	Write short note on 3D rotation.
b.	Discuss any two 3D display methods.

6. Attempt any *one* part of the following:**07 x 1 = 7**

a.	Discuss Bezier Surfaces in detail.
b.	Write short note on Bezier Curves Explicit Formulation.

7. Attempt any *one* part of the following:**07 x 1 = 7**

a.	What do you mean by diffuse reflection and specular reflection? Discuss both in detail.
b.	Discuss Object Space Method and Image Space Method.